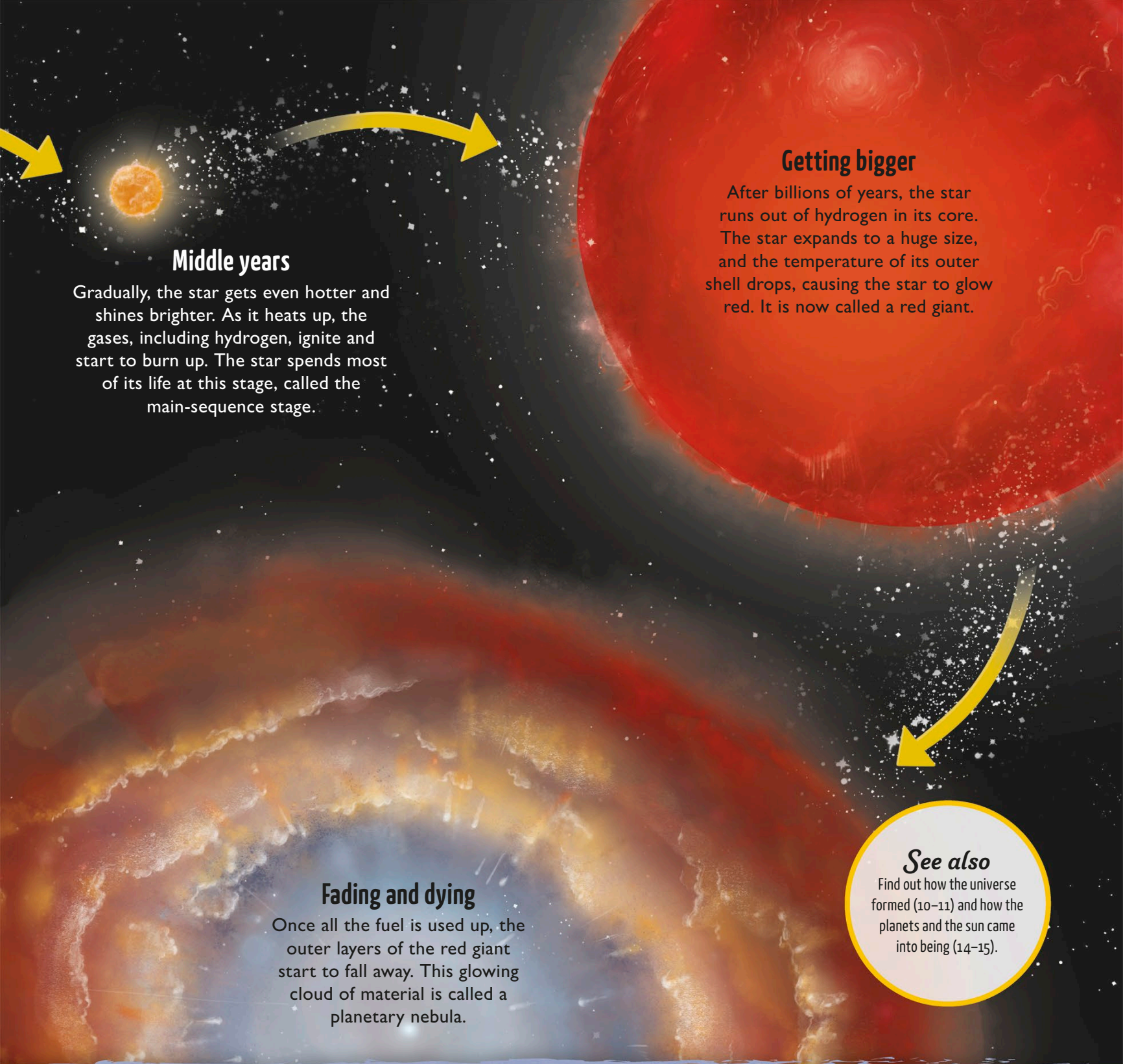




Middle years

Gradually, the star gets even hotter and shines brighter. As it heats up, the gases, including hydrogen, ignite and start to burn up. The star spends most of its life at this stage, called the main-sequence stage.

Getting bigger

After billions of years, the star runs out of hydrogen in its core. The star expands to a huge size, and the temperature of its outer shell drops, causing the star to glow red. It is now called a red giant.

Fading and dying

Once all the fuel is used up, the outer layers of the red giant start to fall away. This glowing cloud of material is called a planetary nebula.

See also

Find out how the universe formed (10–11) and how the planets and the sun came into being (14–15).

Heating up A star's color depends on how hot it is. The hottest stars produce a blue light. Cooler stars have a reddish-orange light. It is possible to see these differently colored stars if you look at the sky through binoculars on a clear night.

